

IoT Sensor-Based Simulation

Temperature Sensor Controlled LED using Arduino (Tinkercad)

Platform	Tinkercad Circuits (Online Simulator)
Microcontroller	Arduino Uno R3
Sensor Used	TMP36 Analog Temperature Sensor
Output Device	LED + 220Ω Resistor
Programming Language	C++ (Arduino IDE)

1. Project Overview

This project simulates an IoT-based temperature monitoring system using Tinkercad Circuits — a free online simulator that requires no physical hardware. An Arduino Uno R3 reads the analog output of a TMP36 temperature sensor, converts it to degrees Celsius, and controls an LED based on a defined temperature threshold. When the temperature exceeds 30°C, the LED turns ON; otherwise, it remains OFF. The Serial Monitor displays real-time temperature readings throughout the simulation.

2. Uses and Real-World Applications

- **Industrial Safety Monitoring:** Factories use temperature sensors to automatically shut down machinery when heat exceeds safe operating limits, preventing equipment damage and fire hazards.
- **Cold Chain & Food Storage:** Refrigerators and freezers in supermarkets and pharmaceutical storage trigger alerts when the internal temperature rises above safe thresholds, protecting perishable goods and medicines.
- **Smart Home Thermostat:** Air conditioners and heaters automatically switch on or off based on ambient room temperature, improving energy efficiency and comfort.
- **Server Room Cooling Systems:** Data centers monitor rack temperatures and activate cooling fans or raise alarms when servers generate excessive heat, protecting expensive hardware from thermal damage.
- **Medical Equipment & Vaccine Storage:** Hospitals and clinics use temperature-controlled cabinets for storing vaccines and blood samples, with automatic alerts if conditions deviate from the required range.
- **Greenhouse & Agriculture Automation:** Automated systems in greenhouses monitor soil and air temperature, triggering irrigation, ventilation, or heating systems to maintain optimal growing conditions.

3. Working Principle

The system follows the fundamental IoT loop: **Sense** → **Process** → **Actuate**.

Step	Description
Sensing	The TMP36 sensor detects ambient temperature and outputs a proportional analog voltage. At 0°C
Signal Conversion	The Arduino reads the analog voltage on pin A0 using its 10-bit ADC (Analog-to-Digital Converter),
Processing	The raw ADC value is converted to voltage using the formula: Voltage = raw × (5.0 / 1023.0). Temp
Decision Making	The computed temperature is compared against a threshold (30°C by default). This is the "intelligen
Actuation	If the temperature exceeds the threshold, the Arduino sends a HIGH signal to pin 13, turning the LE
Monitoring	Temperature readings are continuously printed to the Serial Monitor every second, simulating real-ti

4. Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Microcontroller — reads sensor, runs logic, controls LED
TMP36 Temperature Sensor	1	Analog sensor — outputs voltage proportional to temperature
LED (Red)	1	Visual output — lights up when temperature exceeds threshold
Resistor (220 Ohm)	1	Current limiter — protects LED from burning out
Breadboard	1	Connection platform — no soldering required
Jumper Wires	6+	Electrical connections between components

5. Code Explanation

```
const int sensorPin = A0;
const int ledPin = 13;
float threshold = 30.0; // degrees C

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(sensorPin);
  float voltage = raw * (5.0 / 1023.0);
  float tempC = (voltage - 0.5) * 100.0;

  Serial.print("Temp: ");
  Serial.print(tempC);
  Serial.println(" C");

  if (tempC > threshold) {
    digitalWrite(ledPin, HIGH); // LED ON
  } else {
    digitalWrite(ledPin, LOW); // LED OFF
  }
  delay(1000);
}
```

Variable Declarations (Lines 1–3)

Three variables are declared globally. **sensorPin** (A0) is the analog input pin connected to the TMP36 middle leg. **ledPin** (13) is the digital output pin controlling the LED. **threshold** holds the temperature value (30.0°C) that triggers the LED — this can be changed to test different scenarios.

setup() Function (Lines 5–8)

pinMode(ledPin, OUTPUT) configures pin 13 as an output so it can source current to the LED. **Serial.begin(9600)** initialises the Serial Monitor at 9600 baud rate, enabling real-time temperature data to be printed during simulation.

analogRead and Voltage Conversion (Lines 11–13)

analogRead(sensorPin) reads the raw ADC value (0–1023) from the TMP36's output pin. This is then multiplied by (5.0 / 1023.0) to convert the 10-bit value back into a voltage in the range 0–5 V. This step is necessary because the Arduino ADC maps 0 V to 0 and 5 V to 1023.

Temperature Conversion (Line 13)

The TMP36 datasheet specifies that the sensor outputs 500 mV at 0°C and increases by 10 mV per degree. So subtracting 0.5 V (the 0°C offset) and multiplying by 100 converts voltage directly to Celsius. For example, if voltage = 0.78 V, then tempC = (0.78 - 0.5) x 100 = 28°C.

Serial Monitor Output (Lines 15–17)

Serial.print() and **Serial.println()** send the temperature reading to the Serial Monitor in the format 'Temp: 28.20 C'. This simulates a real IoT dashboard displaying live sensor data.

Threshold Logic / Decision Making (Lines 19–23)

An if-else block compares the computed temperature to the threshold. If tempC exceeds 30°C, **digitalWrite(ledPin, HIGH)** sends 5 V to pin 13, turning the LED ON. Otherwise, **digitalWrite(ledPin, LOW)** sends 0 V, turning it OFF. This is the core automation logic.

delay(1000) — Line 24

Pauses execution for 1000 milliseconds (1 second) before the loop repeats. This controls the sampling frequency of the sensor — one reading per second — preventing the Serial Monitor from being flooded with data.

6. Simulation Screenshots

The following screenshots were taken during the Tinkercad simulation, demonstrating the system behaviour at different temperature values:

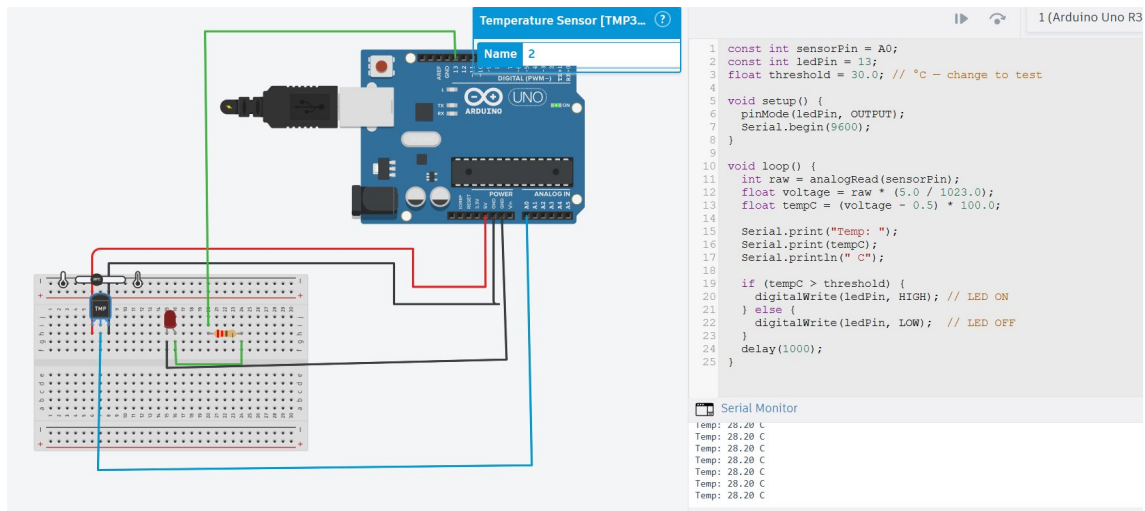


Fig 1 — Simulation running at 28.20°C (below threshold) — LED is OFF

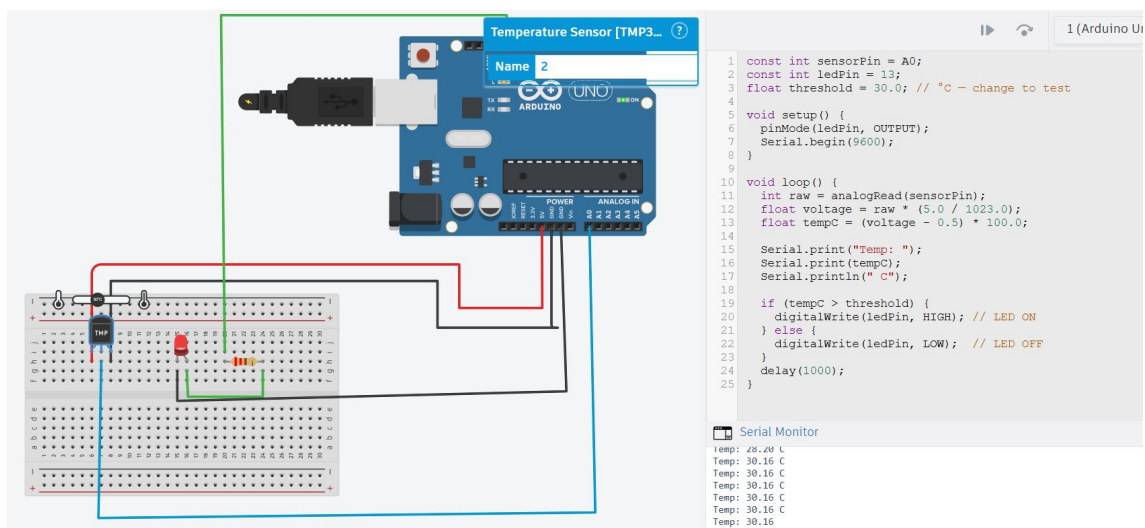


Fig 2 — Temperature raised to 30.16°C (above threshold) — LED turns ON

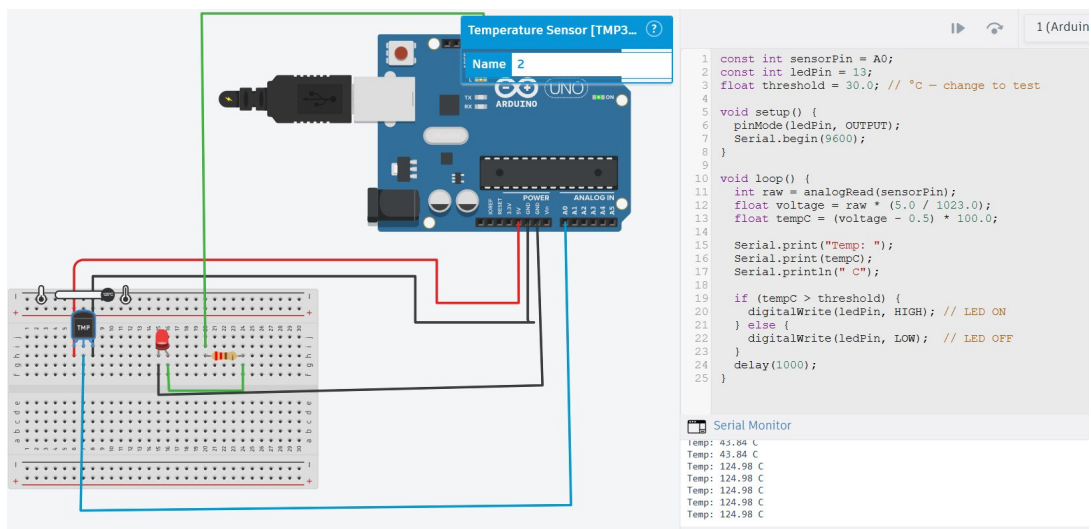


Fig 3 — Temperature increased to 124.98°C — LED remains ON

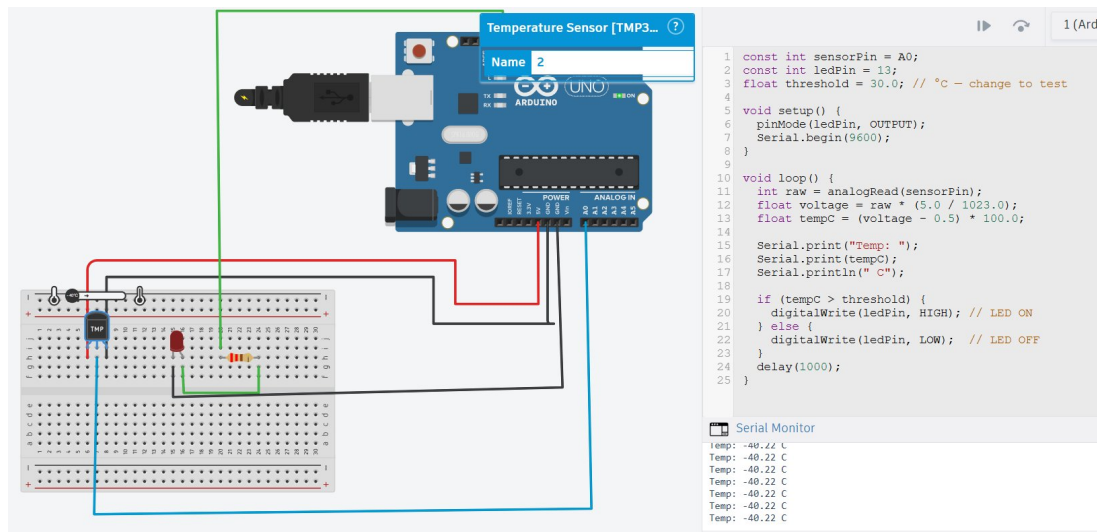


Fig 4 — Temperature set to -40.22°C (minimum) — LED is OFF

7. Conclusion

This Tinkercad simulation successfully demonstrates the core principles of an IoT sensor-based system: sensing physical conditions, processing data in a microcontroller, and actuating an output device based on defined logic. The TMP36 sensor accurately reports temperatures ranging from -40°C to over 125°C, and the Arduino correctly triggers the LED whenever the temperature crosses the 30°C threshold.

The project illustrates how the same logic scales to real-world IoT applications — from smart thermostats to industrial safety systems — where sensing, decision-making, and actuation form the backbone of automation. The Serial Monitor output further simulates remote data monitoring, a key feature of modern IoT deployments.